

Data Shapley for Classification

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Abstract

This project investigates the Data Shapley framework for quantifying the contribution of individual training samples in supervised classification. Through k-NN classification and permutation-based Shapley estimation, we study how Shapley values shift when data are affected by label noise, feature noise, or expanded into higher-dimensional spaces. Experiments on the Wine dataset and synthetic datasets reveal consistent patterns: Shapley values strongly penalize harmful samples, remain stable under noisy features, and degrade smoothly as dimensionality increases.

Declaration

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1 Introduction

Modern machine learning systems rely heavily on the quality and composition of their training data. In many real-world scenarios, datasets are noisy, heterogeneous, or collected from multiple sources with varying reliability. Some samples may be mislabeled, corrupted, or simply uninformative; others may be crucial for shaping the decision boundary. Understanding which data points truly matter is therefore essential for

building robust models, improving data collection pipelines, and diagnosing harmful or low-quality samples.

Traditional evaluation metrics focus exclusively on the final model, providing no insight into the role played by individual training samples. Yet in many practical settings, quantifying the value of each datum is essential. In healthcare, for instance, recent studies have shown that Shapley-based approaches are increasingly adopted in clinical prediction tasks, where sample-level interpretability is essential for assessing data reliability and supporting decision-making [2]. This motivates the need for a principled framework capable of assigning an equitable value to each training sample.

Data Shapley [1] addresses this problem by adapting the classical Shapley value from cooperative game theory to supervised learning. The framework is built on three ingredients: a fixed training set $D = \{(x_i, y_i)\}_{i=1}^n$, a learning algorithm A that maps subsets of D to predictors, and a performance metric V that evaluates the predictor on a reference set. The value of a datum i is defined as its expected marginal contribution to the performance when added to any subset of the remaining data. This formulation satisfies three desirable properties: *nullity* (points that never help receive zero value), *symmetry* (equally useful points receive equal value), and *additivity* (values decompose across independent performance components). These axioms uniquely characterize the Shapley value as the correct notion of equitable data valuation.

Computing the exact Shapley value requires summing over all 2^{n-1} subsets of D , which is infeasible for all but very small datasets. To make the method practical, Shapley values are approximated via Monte Carlo permutation sampling: each random permutation of the training set defines an ordering in which points are added one by one, and the marginal gain in performance is recorded. Averaging these gains across many permutations yields an unbiased estimator of the true Shapley value.

In this project, we adopt the standard Shapley estimator (without truncation), which is well suited for small-to-medium datasets and lightweight models such as k-NN. Algorithm 1 summarizes the procedure.

This framework allows us to investigate how individual samples contribute to generalization under controlled perturbations. In particular, we analyze how Shapley values behave when the dataset is affected by label noise, feature noise, or increased dimensionality which are three common sources of difficulty in real-world learning pipelines. By examining Shapley value distributions and their induced removal effects, we gain insight into the robustness of the method and its ability to identify harmful or uninformative samples. To structure our analysis, the remainder of the paper is organized as follows.

Section 2 describes the datasets used in our experiments. We consider the Wine dataset as a real-world baseline and construct synthetic datasets with controlled perturbations, including label noise, feature noise, and increased dimensionality (4d, 16d, 32d).

Section 3 details the methodology. We outline the train/validation/test protocol, the cross-validation procedure for selecting k in k-NN, the Monte Carlo Shapley estimation process, and the removal experiments used to assess the impact of low- and high-value samples.

Algorithm 1 Shapley Estimation

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1: Input: training set  $D$ , learning algorithm  $A$ , performance metric  $V$ , number of
   permutations  $T$ 
2: Initialize  $\phi_i \leftarrow 0$  for all  $i \in D$ 
3: for  $t = 1$  to  $T$  do
4:   Sample a random permutation  $\pi$  of the indices  $\{1, \dots, n\}$ 
5:    $S \leftarrow \emptyset$ 
6:    $v_{\text{prev}} \leftarrow V(S)$ 
7:   for  $j = 1$  to  $n$  do
8:      $S \leftarrow S \cup \{\pi(j)\}$ 
9:      $v_{\text{new}} \leftarrow V(S)$ 
10:     $\phi_{\pi(j)} \leftarrow \phi_{\pi(j)} + (v_{\text{new}} - v_{\text{prev}})$ 
11:     $v_{\text{prev}} \leftarrow v_{\text{new}}$ 
12:   end for
13: end for
14: Output:  $\phi_i/T$  for all  $i \in D$ 
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Section 4 presents the empirical results. We report Shapley value distributions via quantile plots, and removal curves for both real and synthetic datasets, highlighting how Shapley values behave under different noise levels and dimensionality settings.

Section 7 outlines limitations of the current work.

Section 8 concludes the paper by summarizing the main findings.

2 Datasets

2.1 Wine dataset

The Wine dataset is a classic multi-class classification benchmark from the UCI Machine Learning Repository [3]. It contains 178 samples, 13 numerical features, and 3 well-separated classes corresponding to chemical analyses of wines produced by three cultivators in the same region of Italy. The class distribution is moderately balanced (59, 71, and 48 samples), and the dimensionality is sufficiently rich to support meaningful structure while remaining small enough for exhaustive Shapley estimation.

This dataset is particularly suitable for our study for three reasons. First, its limited size makes it compatible with full Monte Carlo Shapley estimation, which would be computationally prohibitive on larger datasets. Second, the data are clean, fully numerical, and free of missing values, allowing us to isolate the behavior of Shapley values without confounding factors such as noise or irregular sampling. Third, the dataset is known to yield strong performance even with simple models such as k-NN, making it an ideal baseline for evaluating how Shapley values reflect sample importance in a stable, low-noise environment.

2.2 Synthetic datasets

Each synthetic dataset contains 200 samples and a fixed set of three informative features, which define the underlying signal used to generate the binary labels. The

remaining features are deliberately constructed as noise variables, allowing us to control the signal-to-noise ratio and the dimensionality of the input space.

We consider three dimensionality settings: 4, 16, and 32 total features. In all cases, the first three dimensions are informative, while the remaining $n_{\text{extra}} = \{1, 13, 29\}$ features are purely noisy. This design enables us to evaluate how Shapley values behave as the proportion of noise increases and as the feature space becomes progressively higher-dimensional (two factors known to significantly affect the stability of learning algorithms and distance-based models such as k-NN).

Labels are generated by forming a weighted linear combination of the informative features, with decreasing weights to simulate varying relevance. The resulting signal is thresholded at zero to produce a balanced binary classification task. Noisy features are sampled independently from a standard normal distribution, ensuring that they do not carry useful information about the label and therefore act as distractors.

The synthetic datasets used in our experiments are:

- **syn_noi_4d**: 200 samples, 4 features (3 informative + 1 noisy)
- **syn_noi_16d**: 200 samples, 16 features (3 informative + 13 noisy)
- **syn_noi_32d**: 200 samples, 32 features (3 informative + 29 noisy)

3 Methodology

Our experimental pipeline consists of four main components: data splitting, model selection, Shapley value estimation, and removal-based evaluation. Each dataset, real or synthetic, is processed under multiple noise conditions to assess the robustness of Shapley values.

3.1 Train - validation - test split

For each dataset and each noise level, we perform a three-way split into training, validation, and test sets using fixed proportions (60%, 20%, 20%). The training set is used to compute Shapley values, the validation set provides the payoff metric for Shapley estimation, and the test set is reserved for evaluating removal experiments.

Label noise is injected only into the training set, ensuring that validation and test performance reflect the true underlying signal. Noise levels follow the configuration:

$$\text{no_noise} = 0.0 \quad \text{low} = 0.05 \quad \text{medium} = 0.15 \quad \text{high} = 0.30.$$

3.2 Model selection

Since Shapley estimation requires repeatedly training the model on subsets of the training data, we adopt a lightweight classifier: k-NN. The number of neighbors k is selected via cross-validation on the (possibly noisy) training set. We evaluate the candidate set

$$\{1, 3, 5, 7, 9, 12, 14\},$$

and choose the value of k that maximizes validation accuracy. This ensures that Shapley values are computed using a model configuration that is well-adapted to each dataset and noise condition.

3.3 Shapley estimation

Shapley values are estimated using full Monte Carlo permutation sampling. For each dataset and noise level, we generate 100 random permutations of the training set. For each permutation, the model is trained incrementally on growing subsets of the training data, and the marginal gain in validation accuracy is recorded.

We use the standard (non-truncated) Monte Carlo estimator for two reasons:

- the datasets are small (200 samples for synthetic data, 178 for Wine),
- k-NN training is computationally inexpensive for small datasets

The payoff metric is always the validation accuracy, ensuring consistency across datasets and noise levels.

3.4 Removal experiment

To evaluate how Shapley values reflect sample importance, we perform removal experiments on the test set. Given the Shapley values computed on the training set, we remove increasing fractions of training samples according to three strategies:

- **lowest-value first**: removes samples with the smallest Shapley values,
- **highest-value first**: removes samples with the largest Shapley values,
- **random**: removes a random subset of samples (averaged over 40 repeats).

The removal fractions considered are:

$$\{0.0, 0.05, 0.1, 0.15, 0.2, 0.3, 0.5, 0.7\}.$$

For each fraction and strategy, we retrain the k-NN classifier on the remaining training samples and evaluate accuracy on the test set. This procedure reveals how removing harmful samples (low Shapley values) or beneficial samples (high Shapley values) affects generalization performance.

4 Results

We evaluate Shapley values using two complementary visualization tools: (i) normalized quantile plots, which summarize the distribution of Shapley values across clean and noisy samples, and (ii) removal curves, which measure how test accuracy changes when subsets of training points are removed according to their Shapley ranking. Together, these tools allow us to assess whether Shapley values correctly identify harmful samples and whether they reflect the contribution of individual points to generalization.

4.1 Quantile plots

For each dataset and noise level, we compute the normalized quantile function of the Shapley values. When label noise is present, we plot two curves: one for clean samples and one for noisy samples. This visualization highlights how Shapley values separate informative points from corrupted ones. Ideally, noisy samples should exhibit consistently lower Shapley values, reflecting their negative contribution to model performance.

4.2 Removal curves

To quantify the impact of Shapley-guided removal, we evaluate test accuracy after removing increasing fractions of training samples. We compare three strategies:

- **lowest-value first**: removes samples with the smallest Shapley values,
- **highest-value first**: removes samples with the largest Shapley values,
- **random**: removes a random subset.

If Shapley values correctly capture sample importance, removing low-value points should improve or maintain accuracy, while removing high-value points should degrade accuracy. The random baseline provides a neutral reference curve between these two extremes.

4.3 Wine dataset

The Wine dataset provides a clean, low-noise baseline. For this dataset, normalized quantile plots reveal how Shapley values respond to increasing levels of label noise. In the noise-free setting, Shapley values form a smooth and predominantly positive distribution, reflecting the clean structure of the dataset. When label noise is injected, noisy samples consistently occupy the lowest quantiles, confirming that Shapley values correctly identify corrupted points.

As noise increases from low to medium and high, the distribution of clean samples begins to shift downward: a growing fraction of originally clean points receives Shapley values close to or below zero. This behavior is expected, since label noise in the training set distorts the decision boundary and reduces the apparent contribution of clean samples. Under high noise, the positive region of the clean curve becomes noticeably compressed, indicating that even informative samples are harder for the model to exploit. In other words, although Shapley values continue to separate noisy from clean samples, a sufficiently high noise level inevitably affects the valuation of non-noisy points as well. Figure 1 summarizes these trends across noise levels.

The removal curves for the Wine dataset further illustrate how Shapley values capture the contribution of individual training samples (Figure 2).

In the noise-free setting, removing samples with the lowest Shapley values has no noticeable impact on test accuracy. In contrast, removing samples with the highest Shapley values does lead to a degradation in accuracy, although the effect remains limited. This is expected, as the dataset is clean and does not contain truly harmful points.

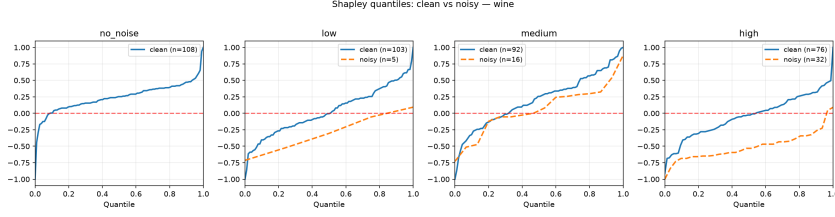


Fig. 1 Normalized Shapley quantiles for the Wine dataset across noise levels.

When label noise is injected, the behavior becomes more pronounced. Across low, medium, and high noise levels, removing low-value samples does not degrade accuracy and often leaves performance essentially unchanged. This indicates that Shapley values correctly identify corrupted or misleading samples: eliminating them does not harm the model and may even stabilize the decision boundary. In contrast, removing high-value samples consistently causes a sharp decline in accuracy, especially under medium and high noise.

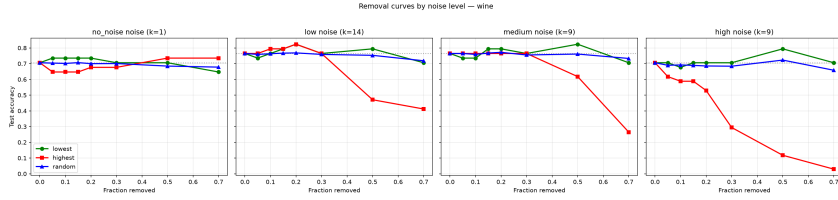


Fig. 2 Removal curves for the Wine dataset.

4.4 Synthetic datasets

The synthetic datasets allow us to analyze how Shapley values behave when both label noise and feature noise increase in a controlled manner. Unlike the Wine dataset, where dimensionality is fixed and all features are informative, the synthetic datasets expand dimensionality by adding features that are explicitly noisy. As a result, higher dimensionality and higher noise are intrinsically coupled: it is not possible to attribute the observed degradation solely to the curse of dimensionality or solely to noise, since both factors grow together. What can be stated clearly, however, is that increasing the number of noisy features reduces the model's ability to exploit the informative ones, and this effect is consistently reflected in the Shapley distributions.

Figures 3–5 show the normalized quantile plots for the 4d, 16d, and 32d configurations. Across all noise levels, noisy samples occupy the lowest quantiles, confirming that Shapley values correctly identify corrupted points. However, the separation between clean and noisy curves progressively shrinks as dimensionality increases. In the 4d dataset, the gap between the two curves is wide and the clean curve rises sharply, indicating that the classifier can easily leverage the informative features. In contrast, the 32d dataset exhibits a much narrower gap: the clean curve becomes flatter and shifts downward, with many clean samples receiving Shapley values close to zero. This

compression reflects the difficulty of k-NN in high-dimensional spaces dominated by noise, where distances become less informative and the marginal utility of individual clean samples decreases.

Removal curves provide additional insight into how Shapley values behave under increasing dimensionality and feature noise (Figures 6–8). In the 4d dataset, the behavior is fully aligned with the expected interpretation of Shapley values: removing samples with the lowest Shapley values leaves test accuracy essentially unchanged, while removing high-value samples produces a clear and immediate degradation. This reflects the fact that, in a low-dimensional setting with few noisy features, the classifier can easily distinguish informative points from uninformative ones.

The situation changes markedly in the 32d dataset. When no label noise is present, the removal of high-value samples can even produce a slight increase in accuracy for small removal fractions, while removing low-value samples may cause a small decrease. This inversion is consistent with the quantile plots: in high-dimensional spaces dominated by noisy features, the separation between clean and noisy Shapley values becomes much weaker, making it harder for the model to identify truly beneficial samples. As a result, removing only a small subset of points does not significantly alter the decision boundary, and the impact of Shapley-guided removal becomes visible only when larger fractions are removed.

When label noise is introduced, the expected pattern re-emerges: removing high-value samples consistently degrades accuracy, while removing low-value samples does not harm performance. However, even in this case, the effect is less pronounced than in the 4d dataset, again reflecting the reduced discriminative power of k-NN in high-dimensional, noise-dominated feature spaces.

Across all noise levels, the 32d dataset exhibits lower overall accuracy than the 4d dataset, confirming that both dimensionality and feature noise jointly reduce the model’s ability to exploit informative samples. These trends demonstrate that Shapley values remain meaningful even under substantial noise, but their sensitivity to sample importance decreases as dimensionality increases.

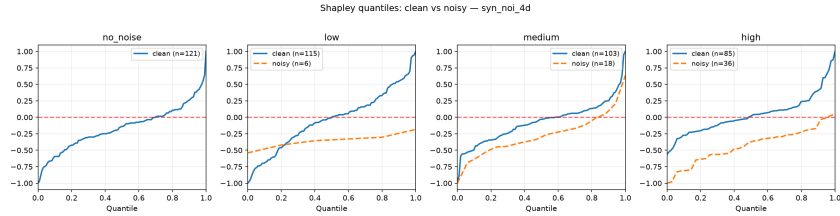


Fig. 3 Shapley quantiles for the synthetic 4d noisy dataset.

5 Limitations

This study adopts the full, non-truncated Monte Carlo estimator for Shapley values. While this choice ensures stable and unbiased estimates on small datasets, it does not scale to larger training sets, where the computational cost becomes prohibitive.

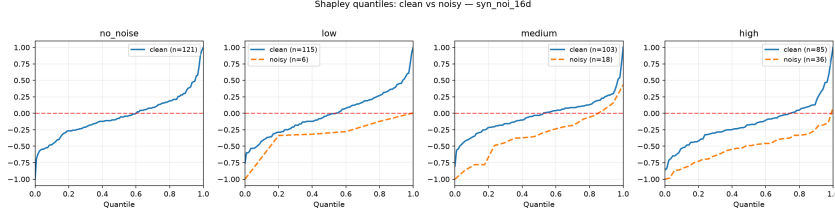


Fig. 4 Shapley quantiles for the synthetic 16d noisy dataset.

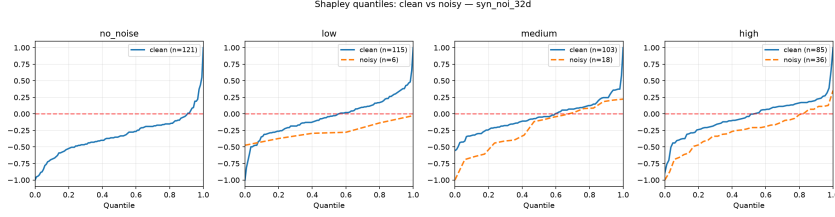


Fig. 5 Shapley quantiles for the synthetic 32d noisy dataset.

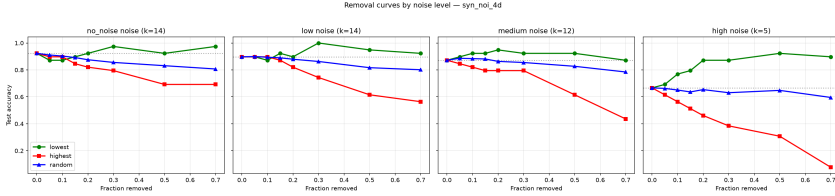


Fig. 6 Removal curves for the synthetic 4d noisy dataset.

Future work could explore more efficient approximations such as TMC-Shapley which reduces the number of required permutations.

A second limitation concerns the choice of classifier. We rely on k-NN because of its simplicity and the absence of training dynamics, which makes Shapley estimation easier to interpret. Extending the analysis to other models such as logistic regression, tree-based methods or small neural networks would provide a broader understanding of how Shapley values behave across different learning paradigms.

A third limitation arises from the structure of the synthetic datasets. In our design, dimensionality increases exclusively through the addition of noisy features. As a consequence, high dimensionality and high noise are intrinsically coupled, making it impossible to isolate the effect of dimensionality alone or noise alone. Future synthetic datasets could include redundant but non-degrading features (e.g., noisy copies of informative dimensions) to study the impact of dimensionality independently from noise. Alternatively, experiments on real high-dimensional datasets would allow evaluating Shapley robustness in settings where dimensionality is naturally large.

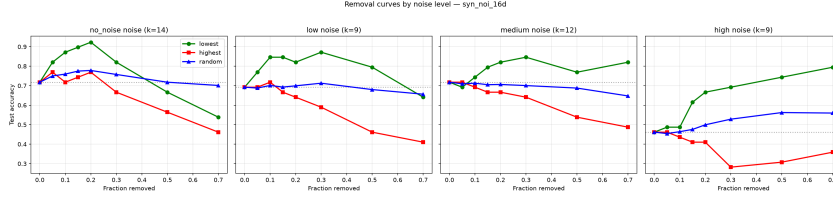


Fig. 7 Removal curves for the synthetic 16d noisy dataset.

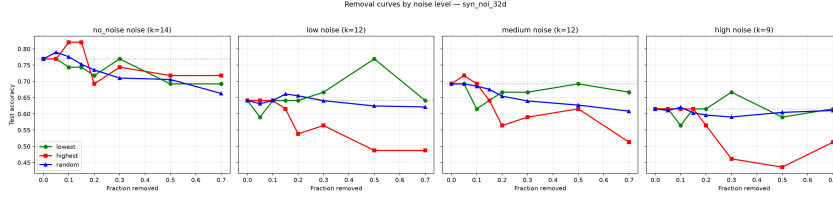


Fig. 8 Removal curves for the synthetic 32d noisy dataset.

6 Conclusion

This study examined the behavior of Data Shapley under controlled perturbations affecting the training data, including label noise, feature noise, and increased dimensionality. Using k -NN classification and full Monte Carlo Shapley estimation, we evaluated how individual samples contribute to generalization and how these contributions shift under different sources of difficulty. Overall, our results show that Data Shapley provides a robust and interpretable measure of sample importance across a variety of noise conditions. The method successfully identifies harmful samples, remains stable under feature-level corruption, and degrades smoothly as dimensionality increases. These findings support the use of Shapley values as a practical tool for data quality analysis, sample-level interpretability, and dataset diagnostics in supervised learning pipelines, especially in settings where understanding the role of individual training samples is essential.

References

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